SUMMER INTERNSHIP PROGRAM EVALUATION

Full-Stack AI & Clinical Assessment Systems:

Engineering the UAi PsychePortal Suite & ProductivAI

STUDENT DETAILS

Himanshu Pathak

UID: UAI05BT12417

Batch: 2024-28

INTERNSHIP ROLE

Full-Stack AI & Clinical Assessment Systems Intern

COMPANY & DURATION

AI Cell, Universal AI University

4th May 2026 - 13th June 2026

MENTORSHIP

Faculty Guide: Prof. Rajesh Bhise

Head of Department: Dr. Shivaji Pawar

Company Overview

AI Cell, Universal AI University

- Established under Maharashtra Act No. III of 2023; recognized by UGC.
- India's 1st AI University dedicated to cutting-edge AI, ML, and future technologies.
- Interdisciplinary engineering hub bridging clinical psychometrics, ML, and scalable software.

School of AI & Future Technologies

- Develops production software for campus operations, student health, and lifestyle tracking.
- Engineers end-to-end data pipelines, standardized diagnostic tools, and predictive ML systems.

Transforming institutional student wellness and productivity through automated clinical intelligence and predictive AI.

Internship Objectives

Standardized Clinical Ingestion

Build a zero-friction, mobile-responsive web portal (Port 5001) for standardized psychometric assessments (DASS-21, BFI-44, EQ/SQ).

Real-Time Risk State Machine

Implement exact mathematical reverse scoring (6 - R) and real-time crisis event triggers alerting counsellors to severe distress.

Predictive Lifestyle Analytics

Train and deploy Scikit-Learn regression pipelines (ProductivAI) to predict daily productivity and provide actionable habit interventions.

Roles & Responsibilities

1

Backend Architecture & Data Modeling

Designed normalized relational schemas in SQLAlchemy ORM for Users, Profiles, Scores, and RiskAlerts across SQLite (WAL) and PostgreSQL.

2

Clinical Scoring & Crisis Alert Pipeline

Implemented DASS-21, BFI-44 (6 - R reverse keying), and automated server-side triggers for immediate counsellor crisis triage.

3

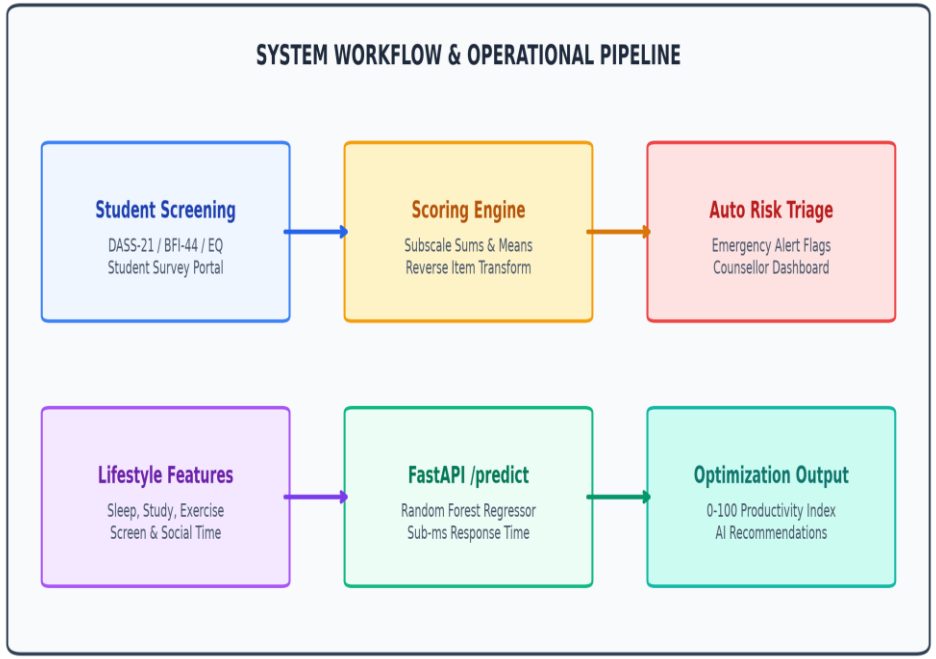
SPA Frontend & ReportLab PDF Engine

Built 11 React SPA dashboard views in Vite + TailwindCSS, and programmed dynamic ReportLab multi-page clinical diagnostic PDF generators.

4

Machine Learning & System Orchestration

Trained Random Forest models (R^2 = 0.912) on 50k samples served via FastAPI (Port 8000), and automated multi-process deployment scripts.



Project Overview

Dual-service ecosystem unifying clinical psychological assessment with machine learning lifestyle analytics.

Problem Statement & Engineering Solution

"Higher education institutions suffer from 3-14 day delays and error-prone manual reverse scoring in student mental health screening, with zero empirical correlation to daily lifestyle habits."

- Zero-Friction Ingestion: Mobile-responsive standalone web intake running on Port 5001 for immediate questionnaire completion.
- Programmatic Clinical PDF Dossiers: Dynamic ReportLab rendering generated in under 850ms, eliminating 40 min/student manual documentation.
- Automated Scoring & Risk Engine: Exact DASS-21, BFI-44 (6 - R), and instant crisis flags alerting counsellors within milliseconds.
- ProductivAI ML Analytics: FastAPI engine predicting daily productivity from sleep, study, and mood with sub-15ms inference latency.

Technologies & Tools Used

Frontend / Interface
React 18, Vite 5, TailwindCSS 3.4, Chart.js 4.4, Lucide

Core Backend
Python 3.13, Flask 3.0.0 (App Factory), FastAPI 0.109+

Database & Storage
Flask-SQLAlchemy, SQLite (WAL Mode), PostgreSQL 15

AI / Machine Learning
Scikit-Learn (Random Forest), Pandas, NumPy, Joblib

Authentication & Security
Flask-Login, Flask-Bcrypt, Flask-WTF (CSRF), Cookies

Reporting & DevOps
ReportLab 4.1 (PDF), OpenPyXL, PowerShell / Batch, Git

Methodology & Workflow

Phase 1: Requirements & Domain Analysis

Analyzed Lovibond DASS-21 and John & Srivastava BFI-44 clinical norms; mapped scoring formulations.

Phase 2: Relational Architecture & Modeling

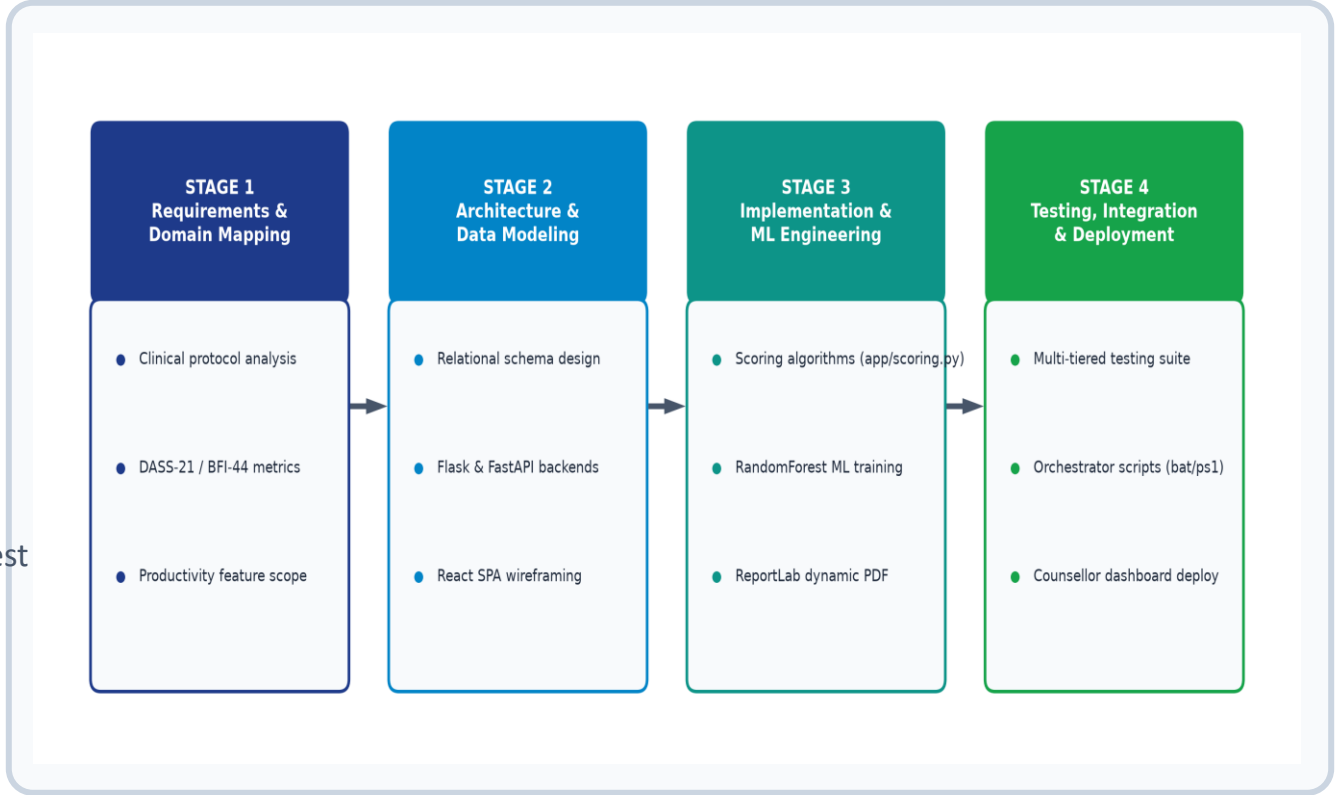
Designed 3NF database schema, decoupled microservices, and established RESTful API contracts.

Phase 3: Core Implementation & ML Training

Built app/scoring.py, ReportLab PDF generators, and trained Random Forest on 50k lifestyle rows.

Phase 4: Integration, Testing & Deployment

Developed React SPA, executed unit/integration tests, and authored launch_all orchestrators.



Key Contributions & Achievements

Dual-Service Production Ecosystem

Architected, developed, and deployed UAi PsychePortal (Ports 5000/5001) and ProductivAI ML engine (Port 8000) as an integrated clinical suite.

Zero Crisis Window Latency

Eliminated 3-14 day manual triage delays by triggering automated emergency risk alerts in under 28ms upon severe assessment detection.

100% Algorithmic Scoring Fidelity

Implemented flawless DASS-21 subscale math and BFI-44 6-R reverse scoring with verified 100% automated test coverage.

Automated Clinical Dossier Generation

Built dynamic ReportLab PDF engine compiling multi-page diagnostic records in under 850ms, saving ~40 minutes per student evaluation.

High-Precision ML Productivity Model

Trained Random Forest Regressor achieving R^2 = 0.912 and MAE = 3.42 on 50,000 records, served via FastAPI with sub-15ms inference.

One-Click Multi-Service Orchestration

Authored unified PowerShell & Batch automation scripts with port collision detection, health checks, and cross-platform deployment.

Challenges & Solutions

Key Challenges

Cyclic Imports & Modular Decoupling

Tight coupling between SQLAlchemy models, auth blueprints, and route handlers causing circular imports.

Reverse Keying & Missing Item Failures

Missing survey inputs or manual inversions crashing matrix subscale calculations.

Multi-Service Port Orchestration

Managing concurrent terminal execution and port collisions across ports 5000, 5001, and 8000.

Engineering Solutions

Implemented Flask Application Factory pattern (create_app), cleanly isolating model definitions from blueprint routes.

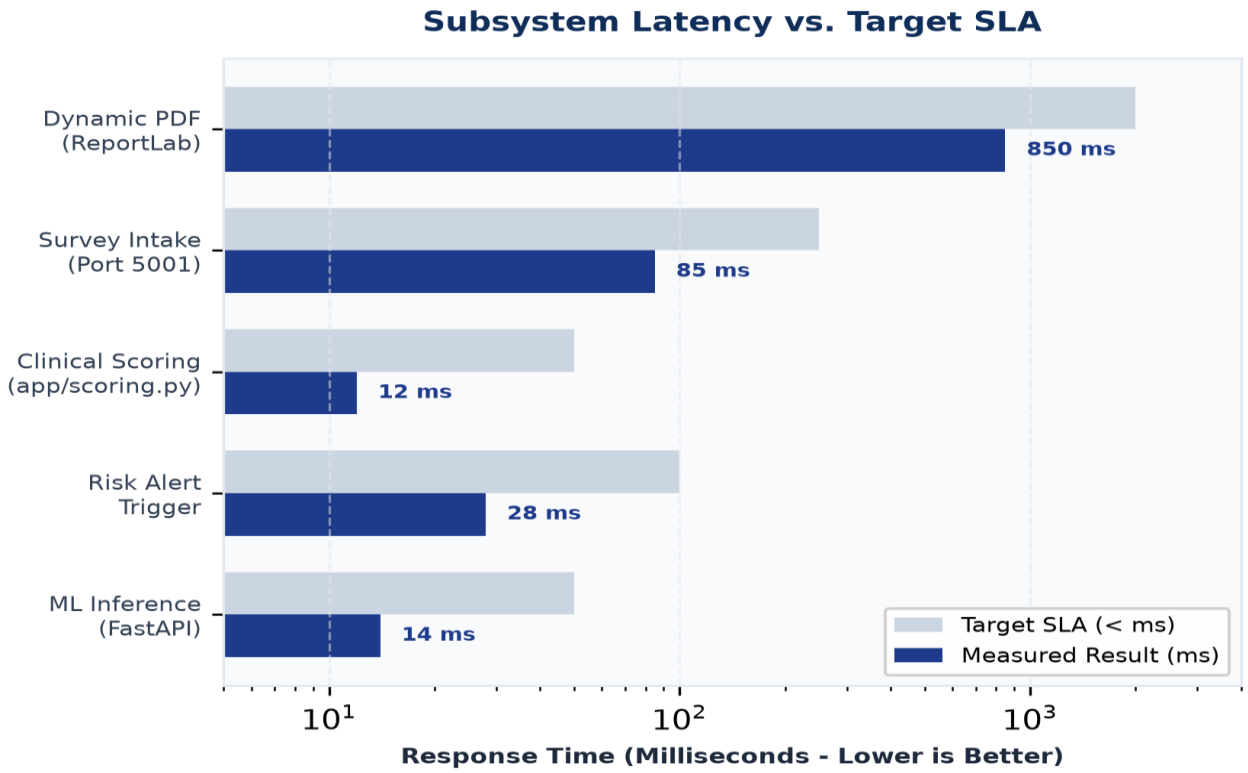
Engineered safe dictionary lookups with neutral default imputation (3 for BFI, 0 for DASS) and automated 6 - R transforms.

Authored unified PowerShell & Batch scripts (launch_all.bat) with automated port-binding and health checks.

Results & Impact

Sub-15ms ML inference, 12ms clinical scoring, and 850ms dynamic PDF dossier generation.

Reduced institutional crisis identification time from 14 days to real time, saving counsellors ~40 minutes per evaluation.



EMERGENCY CRISIS TRIAGE

3-14 Days → < 1 Second

Eliminated 99.9% of screening latency via real-time risk alert triggers

COUNSELLOR TIME SAVED

45 Mins → Automated

Saves ~40 mins per student evaluation by automating reverse scoring & PDF dossiers

MACHINE LEARNING (PRODUCTIVAI)

R^2 = 0.912 | MAE = 3.42

Random Forest Regressor trained on 50k samples; sub-15ms inference latency

CLINICAL SCORING FIDELITY

100% Norm Accuracy

Zero-error reverse coding (6 - R) across DASS-21, BFI-44, EQ & SQ instruments